

Team turn — 2026-07-03 next. The down-ladder is now a $d(\nu)$ question. Compute it.

SOD check first. The gate sharpened this turn — here's the clean state.

Where we are

- The down-ladder gate is now SHARP. Quark mass = $d(\nu)$ [deposit density] $\times$ k_s [cohomology], both forced. Grace computed her half: $k_s = \dim \ker R$ is small — $\{2, 4, 6, 8\}$ ($S^4 \rightarrow 2$, wall-flag $\rightarrow 4$, $Q^5 \rightarrow 6$, color-Fock $\rightarrow 8$), none near 20. That's a *useful computed narrowing*: since the ladder is $\{1, 20, 900\}$ and k_s is small, the large numbers (20, 900) must come from Lyra's $d(\nu)$ — the Harish-Chandra formal degree. The question “why 20, why 900” moved from geometry to rep-theory, with a definite answer from Grace's side.
- Confinement = “commit 27, read out 1” (Lyra F459): the 3-quark system commits the full $N_c^3=27$ color tensor ($3 \otimes 3 \otimes 3 = 10 \oplus 8 \oplus 8 \oplus 1$); only the singlet $\Lambda^3=1$ manifests; the other 26 are committed-but-confined. Unifies committed-info + Pauli(exterior algebra) + confinement — same N_c^3 as the α count. (Algebra is Günaydin-Gürsey — cite, don't claim; ours is the commit-and-read language.)
- Running vs dressing disentangled (K650, Elie 4553): the 1-2% cross-sector residual is QCD running (cancels same-sector: $m_s/m_d=20$ is 0σ); membrane dressing is the absolute constituent addition $\sim m_p/N_c$. Layer-2 dressing is a constituent-mass prediction with running subtracted — NOT a ratio residual.
- Count 8.

The frontier — COMPUTE $d(\nu)$ (Lyra, highest-leverage)

The whole down sector's derived-strong test is now one rep-theory computation:

Compute $d(\nu)$ (the Harish-Chandra formal degree) at the three generation K-types, reconcile with Grace's $k_s \in \{2,4,6,8\}$, and read $d(\nu) \times k_s$ against the bare ladder $N_c^2 \times \{1, 20, 900\}$.

- COMPUTE, do NOT fit. Grace held the $\chi=4$ line (it's the honest wall-flag Betti number, not a fish); Lyra holds the $d(\nu)$ side the same way. If $d(\nu) \times k_s$ forces $\{1, 20, 900\}$, the bare down sector is derived-strong. If not, honest negative — the ladder is a lead, not a bank.
- Recall $d(5/2)=0$ (massless bare gen-1) — the formal-degree zeros are load-bearing; work them carefully.

Per-CI

- LYRA (primary): compute $d(\nu)$ at the three K-types $\rightarrow$ the ladder gate. Reconcile with Grace's k_s . Solo: close the ν -scale coefficients + finish registry backfill (fresh reads). Check Elie's ν -regime lead (does singlet retention $\sim 10^{-7}$ = a truncation residue $\sim 2^{\{-3g\}}$?).
- GRACE: k_s half done ($\{2,4,6,8\}$); now geometry-checker on Lyra's $d(\nu)$ reconciliation + the constituent-dressing geometry (Layer 2, where running is already subtracted).
- ELIE: checker on $d(\nu) \times k_s$ (forced vs fit — hold the $\chi=4/12$ fence); the ν -regime residue lead (retention = $2^{\{-3g\}}$); constituent-dressing (K650 rule — dressing lives in the absolute constituent mass, not the ratio residual; top control $f_{had} \approx 0$).
- KEEPER: adjudicate $d(\nu) \times k_s$ (forced vs fit); hold the running-vs-dressing line (K650) — no ratio residual reported as dressing; keep the two-layer + regime frame honest; the residue-family thread (ν , and Λ — see Casey's item). SOD first.
- CAL: post-landing cold-read. α tier-call reserved for Casey.

Disciplines armed

- Compute the object, don't fit the number ($d(\nu)$ and k_s are both forced; their product is forced-or-not).
- Derive BARE; predict dressing = retention $\times$ thickness, running subtracted first.
- Simplest one-line-checkable forms (ad-AI-num armor) $\cdot$ Five-Absence $\cdot$ target-innocence $\cdot \sigma$ scheme-aware $\cdot$ terminal is an answer $\cdot$ pre-audit.

Standing for Casey

$d(\nu)$ close (pending) · two-layer + regime mass model as the mass-sector frame (ratify) · down-row re-tier (re-framed to bare) · σ -metric + cheapness ratification · α tier-call (Wyler-informed) · CLAUDE.md line 0.5.

EOD target

$d(\nu)$ computed at the three K-types $\rightarrow$ the ladder $\{1, 20, 900\}$ forced-or-not (with Grace's small k_s). ν coefficients closed-or-flagged. Backfill advanced. The residue-family question opened ($\nu \rightarrow \Lambda$). Derived-strong grows only if $d(\nu) \times k_s$ forces the powers.

— Keeper, next turn. The ladder is a $d(\nu)$ computation now — Grace narrowed it, Lyra closes it. Compute, don't fit. Mechanism is the prize.